

$\mathbb{Z}_7$ Defect Cosmology

Domain Walls, Phase Transitions, and the Zero-Relic Prediction

UGP Physics Tutorial Series

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Contents

1	Phase Transitions in Cosmology	3
1.1	Water freezing: the everyday analogy	3
1.2	Phase transitions in the early universe	3
1.3	The defect zoo	4
2	The $\mathbb{Z}_7$ Domain-Wall Problem	4
2.1	The Φ_{MDL} field and its seven vacua	4
2.2	The ordering crossover at $T_G \approx 0.70$ GeV	4
2.3	Why domain walls are fatal	5
3	The Kibble Mechanism	6
3.1	Causal horizons in the early universe	6
3.2	How the Kibble foam forms	6
3.3	A magnet analogy	6
4	The GTE Resolution: Coupling-Bias Annihilation	6
4.1	Why a simple symmetry argument fails	7
4.2	The canonical coupling V_{coupling}	7
4.3	The bias magnitude	7
4.4	The collapse: walls accelerate and annihilate	7
4.5	Why the bias is “the theory’s own”	8
5	The Kibble Foam: A TikZ Diagram	8
6	The Falsifiable Prediction	9
6.1	What “zero surviving network” means	9
6.2	The gravitational-wave ceiling	9
6.3	Falsification criteria	10
7	Why This Is Remarkable	10
7.1	The domain-wall problem as a consistency test	10
7.2	Three approaches to resolution	10
7.3	The falsifiable nature of the prediction	11
7.4	Connection to the broader GTE picture	11

A Quick Reference: Timeline and Scales	12
B Additional References	12

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Prerequisites (read first): *The Cosmological Constant* · *Gravity from Description-Length Minimization*

See also: *The Φ_{MDL} Field*

Claim-Strength Taxonomy

Throughout this tutorial, results are labeled with a confidence grade:

- **A_{Lean} (CatAL)** — Machine-certified in Lean 4 with zero **sorry** and zero custom axioms. Independently verifiable by anyone with Lean installed.
- **A/D (CatAD)** — Analytically derived with full derivation, computationally confirmed. Not yet machine-certified.
- **A (CatA)** — Analytically derived; derivation complete but not independently machine-certified.
- **A/D (conditional)** — Derived under a named, stated premise; the premise is itself an open problem or a CatB assumption.
- **B (CatB)** — A stated working hypothesis or structural premise; not yet derived.

1 Phase Transitions in Cosmology

The one-line answer

As the early universe cooled, it underwent phase transitions just like water freezing into ice. When different regions of space independently “chose” their low-temperature configuration, boundaries formed between them — called *defects* or *domain walls*. If stable, they would have been catastrophic for the universe’s development.

1.1 Water freezing: the everyday analogy

When liquid water cools below 0°C, it freezes into ice. In a large lake, the freezing does not happen all at once from a single point; instead, tiny ice crystals nucleate simultaneously at many *independent* locations across the water. Each crystal grows outward, but adjacent crystals were not in contact when they started — they independently chose their crystal orientation. When the growing crystals meet, a boundary called a **grain boundary** forms at their interface. The grain boundary is a defect: a region where the crystal lattice is mismatched.

The same logic applies to the universe at cosmic scales.

1.2 Phase transitions in the early universe

Immediately after the Big Bang, the universe was unimaginably hot. At those temperatures, all the symmetries of the laws of physics were *manifest*: particles and forces behaved in unified, symmetric ways.

As the universe expanded and cooled, it passed through a series of **phase transitions** — moments when the symmetric high-temperature behaviour suddenly gave way to a broken-symmetry low-temperature configuration. The most famous example is *electroweak symmetry breaking* at about 10^{15} K (roughly 100 GeV in energy), when the photon and the weak gauge bosons (W , Z) “split apart” from a unified electroweak field.

Key idea: vacuum selection

A phase transition happens when the system chooses a new *ground state* (lowest-energy configuration), called the *vacuum*. If the new ground state is not unique — if several equally low-energy options exist — then different regions of space independently choose different options. This is the origin of defects.

1.3 The defect zoo

Different types of vacuum degeneracy produce different types of defects:

Defect type	Origin	Dimensionality
Cosmic strings	$U(1)$ vacuum degeneracy (a circle's worth of choices)	1D lines
Domain walls	Discrete vacuum degeneracy (a finite set of choices)	2D sheets
Monopoles	Higher-symmetry breaking	0D points

The GTE framework has a *discrete* vacuum structure: the Φ_{MDL} field can sit in any of seven degenerate ground states, indexed by elements of $\mathbb{Z}_7 = \{0, 1, 2, 3, 4, 5, 6\}$. This leads to domain walls.

2 The $\mathbb{Z}_7$ Domain-Wall Problem

The domain-wall problem in one sentence

The GTE field Φ_{MDL} has seven equally-valid ground states; if the universe randomly populated them during a phase transition and they remained, stable domain walls would have come to dominate the universe's energy budget — causing it to re-collapse. This is a known threat to any theory with multiple discrete vacua.

2.1 The Φ_{MDL} field and its seven vacua

The GTE framework is built around a physical field, written Φ_{MDL} , that permeates all of space. This field obeys a $\mathbb{Z}_7$ -symmetric potential — a potential energy landscape with *seven equally-spaced minima* arranged like seven valleys on a ring.

Vacuum $k = 0$: the physical vacuum. Today the field sits everywhere in the ground state $k = 0$ — the true physical vacuum. Particles are not little balls floating in this field; they are *topological kinks*: regions where the field has permanently wound from one vacuum sector to another. An electron, for example, is a kink with winding number $w = 4$ (the field changes from $k = 0$ to $k = 4$ as you cross the kink from left to right).

Why seven? The number 7 is not chosen by hand. It is selected by the Minimum Description Length (MDL) principle acting on the arithmetic of the Generative Triple Evolution (GTE) sieve: among all prime moduli, $|\mathbb{Z}_7| = 7$ is the unique MDL-minimal choice consistent with the Standard Model's fermion and generation structure. The GTE polynomial $p(L, C, R) = C + R - CR - LCR \pmod{7}$ encodes all of this.

2.2 The ordering crossover at $T_G \approx 0.70$ GeV

At high temperature, thermal fluctuations are large enough to wash out the distinctions between the seven vacua: the $\mathbb{Z}_7$ symmetry is *restored*. As the universe cools, there is a critical temperature below which the $\mathbb{Z}_7$ order re-establishes itself.

For the Φ_{MDL} field, the relevant criterion is *Debye–Waller melting*. The $\mathbb{Z}_7$ order melts when thermal fluctuations exceed the spacing between vacua. Setting the Debye–Waller condition, $\langle\varphi^2\rangle_T = T^2/12$, the ordering-crossover temperature is:

The GTE ordering-crossover temperature

$$T_G \approx 0.70 \text{ GeV} \quad (\text{normalization bracket } 0.70\text{--}1.24 \text{ GeV})$$

This temperature lies between the QCD scale ($\sim 0.2 \text{ GeV}$) and the electroweak scale ($\sim 100 \text{ GeV}$). It defines a new epoch in the GTE thermal history.

As the universe cools through T_G , the $\mathbb{Z}_7$ vacuum is “re-made” locally in each causally disconnected region of space independently.

2.3 Why domain walls are fatal

If different regions independently chose different vacua, and if no mechanism existed to distinguish them, the following would happen:

1. A patchwork of domains forms, each in one of the seven vacua.
2. The boundaries between adjacent domains are domain walls — 2D sheets of elevated potential energy.
3. A domain-wall network has an energy density $\rho_{\text{wall}} \propto \sigma/t$, where σ is the wall tension and t is time.
4. In a flat universe, ρ_{wall} decays *more slowly* than matter density ($\rho_m \propto t^{-2}$) or radiation density ($\rho_r \propto t^{-8/3}$).
5. Eventually, ρ_{wall} would come to dominate the total energy.

For the Φ_{MDL} field, the wall tension is fixed by the BPS kink integral:

$$\sigma = \frac{8}{49} m_\tau f^2 = 0.29010 \text{ GeV}^3 \quad (\text{canonical normalization}).$$

An unbiased scaling network would come to dominate the energy budget at

$$z_{\text{dom}} \approx 832 \quad (T_{\text{dom}} \approx 0.196 \text{ eV}),$$

exactly at the epoch of recombination ($z \approx 1100$). This would violate the ZKO (Zel’dovich–Kobzarev–Okun) anisotropy bound on CMB temperature variations by a factor of $\approx 3.4 \times 10^3$: catastrophically too large.

The catastrophe, quantified

An unbiased $\mathbb{Z}_7$ domain-wall network would violate the CMB anisotropy bound by a factor of ≈ 3400 at recombination. Every theory with discrete vacuum degeneracy must solve this problem. This is the *domain-wall problem*.

3 The Kibble Mechanism

Key idea

The Kibble mechanism (Tom Kibble, 1976) explains why domain walls form whenever a theory has multiple degenerate ground states and the universe undergoes a phase transition. It is a general causality argument: regions of space that were not in causal contact at the phase transition could not have coordinated their vacuum choice.

3.1 Causal horizons in the early universe

In the early universe, space was expanding so rapidly that distant regions had never been in causal contact (no light signal could have traveled between them). The **Hubble horizon** at any epoch is the characteristic scale over which regions can communicate: roughly c/H , where H is the Hubble rate.

At the ordering-crossover T_G , the Hubble horizon sets the coherence length:

$$\xi \sim \frac{1}{T_G} \approx \frac{1}{0.70 \text{ GeV}} \approx 3 \times 10^{-16} \text{ m}.$$

Regions separated by more than ξ could not coordinate their vacuum choice — they independently and randomly picked one of the seven vacua.

3.2 How the Kibble foam forms

1. At $T > T_G$: $\mathbb{Z}_7$ symmetry is restored; the field fluctuates uniformly among all sectors.
2. At $T = T_G$: each causal patch independently “freezes” into one of the seven vacua, with probability $\approx 1/7$ per vacuum.
3. Adjacent patches that froze into *different* vacua are now separated by a domain wall.
4. The result is a complex, tangled foam of domain walls permeating all space.

This foam is sometimes called **Kibble foam**. For the $\mathbb{Z}_7$ case, a lattice simulation confirms: all seven vacua are populated at $\approx 1/7$ each, six of the seven labels percolate (infinite walls are generic), and 91% of the wall area separates *adjacent* vacua (neighbouring k values — $|\Delta k| = 1$).

3.3 A magnet analogy

Imagine cooling a ferromagnet below its Curie temperature. Each tiny domain independently aligns its spins along one of the available magnetic directions. Where two domains with opposite spin orientations meet, a domain wall forms. The foam of magnetic domain walls that result is exactly the Kibble foam in three-dimensional form.

In the cosmic case, the “spin directions” are replaced by the seven $\mathbb{Z}_7$ vacua, and the “ferromagnet” is the entire early universe.

4 The GTE Resolution: Coupling-Bias Annihilation

The resolution in one sentence

The Φ_{MDL} Lagrangian contains a canonical coupling term that breaks the exact $\mathbb{Z}_7$ symmetry between the seven vacua, creating a tiny but nonzero energy bias that drives the entire foam to collapse within $\sim 10^{-23}$ s — long before any cosmological damage can occur.

4.1 Why a simple symmetry argument fails

One might hope that the seven vacua, being exactly degenerate, would stay exactly degenerate forever. Inside the pure φ -sector of the Φ_{MDL} field, this is indeed the case: the $\mathbb{Z}_7$ shift symmetry is exact to *all orders* in perturbation theory, and no bias can arise. On its own, the vacuum energy cannot distinguish between $k = 0$, $k = 1$, or any other vacuum label. This means the catastrophe is not automatically avoided.

4.2 The canonical coupling V_{coupling}

The escape is provided by the *full* Lagrangian of the theory. The Φ_{MDL} field does not live alone; it couples to a second field χ (the color sector), and the canonical coupling between them is:

$$V_{\text{coupling}} = \varepsilon \varphi^2 (D_\mu \chi)^2,$$

where $\varepsilon = 7/9$ (machine-certified in Lean 4, zero **sorry**). This coupling is *not* $\mathbb{Z}_7$ -shift invariant.

Why this breaks the symmetry. In vacuum k , the field value is $\varphi_k =$ (the field amplitude in sector k). Because the coupling involves φ^2 , different vacua give different values of φ_k^2 , and thus different kinetic normalizations for the χ field:

$$Z_k = 1 + 2\varepsilon \varphi_k^2, \quad k = 0, 1, \dots, 6.$$

The vacuum $k = 0$ gives $Z_0 = 1$ (since $\varphi_0 = 0$ in the physical vacuum); the other vacua give $Z_k > 1$. This difference in kinetic normalization means the free energies of the seven vacua are *no longer equal* once the color sector is thermally populated.

4.3 The bias magnitude

The vacuum-label splitting, computed from the renormalization-group-improved effective potential anchored at the GTE EFT boundary Λ_{GTE} , is:

$$|\Delta V(0 \rightarrow 1)|(T_G) \approx 3.8 \times 10^{-2} \text{ GeV}^4 \quad (\text{systematic battery: } 5 \times 10^{-4} - 2 \times 10^{-1}), \quad (1)$$

of which $+2.8 \times 10^{-2} \text{ GeV}^4$ is a temperature-independent Coleman–Weinberg component. The annihilation does not even require the thermal window; the $T = 0$ contribution alone is sufficient.

Key numbers

Quantity	Value	Notes
Wall tension σ	0.29010 GeV^3	From BPS kink integral
Bias $ \Delta V $	$3.8 \times 10^{-2} \text{ GeV}^4$	RG-improved at T_G
Critical radius R_c	7.7 GeV^{-1}	Above this, walls collapse
Foam lifetime t_{wall}	$\approx 5 \times 10^{-24} \text{ s}$	Full collapse time
Annihilation temp. T_{ann}	$\approx T_G \approx 0.70 \text{ GeV}$	Same epoch as formation
Margin before BBN	$\approx 700\times$	Gone 700 times before nucleosynthesis

4.4 The collapse: walls accelerate and annihilate

Once the bias $|\Delta V|$ exists between adjacent vacua, a wall segment feels a *pressure difference* from either side. The domain in the energetically preferred vacuum ($k = 0$) exerts more pressure than

the other side. As long as the wall’s curvature radius exceeds the critical value $R_c = \sigma/\Delta V \approx 7.7 \text{ GeV}^{-1}$, the wall accelerates inward, converting the less-favored vacuum into the favored one.

The entire Kibble foam collapses with a characteristic time:

$$t_{\text{wall}} \approx 5 \times 10^{-24} \text{ s} = 8 \times 10^{-18} H^{-1}(T_G).$$

This is about $\sim 10^{-23} \text{ s}$ total — the universe was fully “re-unified” into the $k = 0$ vacuum within about one hundred-trillionth of a trillionth of a second.

The timeline. Big-Bang nucleosynthesis (BBN) — the epoch when the lightest nuclei (hydrogen, helium, lithium) formed — occurs at $T_{\text{BBN}} \approx 1 \text{ MeV}$. The domain walls annihilated at $T_{\text{ann}} \approx 0.70 \text{ GeV} \approx 700 T_{\text{BBN}}$: they were gone 700× before the protons and neutrons even began to fuse.

4.5 Why the bias is “the theory’s own”

This is the crucial structural point. The coupling V_{coupling} is not added to the theory to solve the domain-wall problem. It is the *canonical* coupling between the φ and χ sectors forced by the MDL uniqueness of the Φ_{MDL} Lagrangian. Its coefficient $\varepsilon = 7/9$ is derived, not fitted. The resolution of the domain-wall problem is therefore *automatic*: the same coupling that is required for the particle spectrum and color sector simultaneously annihilates the cosmological defect foam.

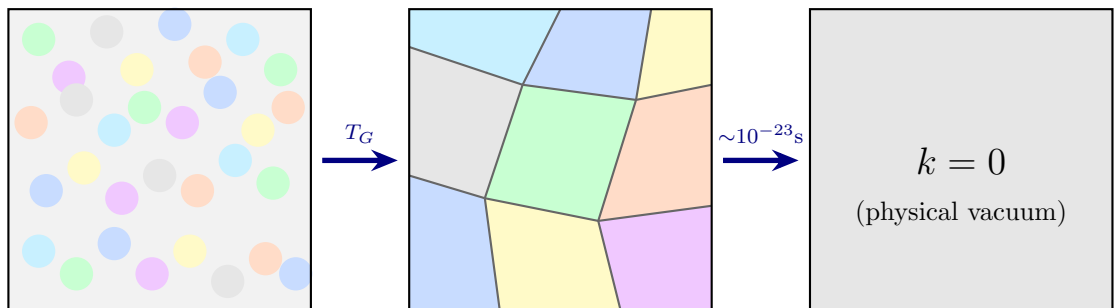
Cross-level consistency remark

At Level 1 (algebraic certificate), the CA fixed-point theorem establishes that $k = 0$ is the unique fixed point of the p -dynamics at every ring size (vacuum_unique_temporal_fixed_point_ring, **CatAL**). At Level 2 (continuum field), the RG-improved effective potential also selects $k = 0$ across the full systematic battery. The algebraic structure and the thermodynamic bias independently converge on the same vacuum.

5 The Kibble Foam: A TikZ Diagram

The sequence below illustrates the formation and annihilation of the Kibble foam.

- (1) High temperature, $T \gg T_G$ (2) Kibble foam, $T \lesssim T_G$ (3) After annihilation, $T_{\text{ann}} \approx T_G$



$\mathbb{Z}_7$ symmetry restored
field fluctuates freely

each causal patch freezes
independently into one of 7 vacua

coupling bias selects $k = 0$
zero surviving defect network

Vacua:



$k = 0$



$k = 1$



$k = 2$



$k = 3$



$k = 4$



$k = 5$



$k = 6$

In stage (2), the thick black lines are domain walls — boundaries between regions that froze into different vacua. In stage (3), the coupling bias has driven all domains to collapse into $k = 0$; the walls have annihilated. This happens within $\sim 10^{-23}$ s, far before any cosmological epoch of interest.

6 The Falsifiable Prediction

Key idea

Because the wall tension σ and the annihilation temperature T_{ann} are both *parameter-free* (derived from first principles with no free parameters fitted), the relic inventory is sharp: zero surviving defect network, with a specific gravitational-wave ceiling and a specific neutrino-sector prediction. These are falsifiable: any confirmed domain-wall relic would rule out the GTE defect-cosmology mechanism.

6.1 What “zero surviving network” means

The zero-relic prediction means:

- **No domain walls survive** to Big-Bang nucleosynthesis (BBN), to the epoch of recombination, or to today.
- **No stochastic gravitational-wave background** from domain walls: the foam never reaches the scaling regime in which it would radiate gravitational waves at observable levels.
- **No extra relativistic species**: the transient foam carries at most 4.2% of the radiation density for only $\sim 10^{-22}$ s before rethermalizing into Standard-Model quanta, giving $\Delta N_{\text{eff}} \approx 0$ — well within current observational bounds.

6.2 The gravitational-wave ceiling

The most precisely quantified prediction is the **GW ceiling**: the upper bound on the stochastic gravitational-wave background from this domain-wall sector. Because the walls annihilate before entering the scaling regime, the gravitational-wave signal is bounded by the energy density of the transient foam at the moment of annihilation:

Gravitational-wave ceiling (GTE zero-relic prediction)

$$\Omega_{\text{GW}} h^2 \lesssim 3 \times 10^{-47} \quad \text{at} \quad f \approx 0.02 \text{ Hz}$$

This is in the LISA frequency band, but approximately 25 orders of magnitude below LISA’s sensitivity threshold ($\Omega_{\text{GW}} h^2 \sim 10^{-12}$ at that frequency). GTE predicts that *any* common-spectrum stochastic gravitational-wave process seen by pulsar-timing arrays or LISA is **not** from this $\mathbb{Z}_7$ domain-wall sector.

6.3 Falsification criteria

Prediction	Value / bound	How to falsify
Zero relic network	No domain-wall relics to BBN	Detection of a cosmological domain-wall relic at any epoch
GW ceiling	$\Omega_{\text{GW}} h^2 \lesssim 3 \times 10^{-47}$ at 0.02 Hz	Confirmed detection of a GW stochastic background attributed to a $\mathbb{Z}_7$ -type domain-wall sector
ΔN_{eff}	≈ 0 (no extra relativistic species)	CMB-S4 or Simons Observatory precision measurement $\Delta N_{\text{eff}} > 0.1$ attributable to domain walls
Vacuum selected	$k^* = 0$ (unique)	Any observation implying residual $\mathbb{Z}_7$ vacuum degeneracy in the present-day universe

What would *not* be falsified by a domain-wall relic detection: the substrate identification (Φ_{MDL} as the physical field), the particle spectrum derivations, the GTE polynomial, or the broader derivation tower. The detection would falsify specifically the coupling-bias annihilation mechanism and the ordering-crossover epoch, requiring a different resolution of the domain-wall problem.

7 Why This Is Remarkable

The three-sentence summary

Any theory with discrete vacuum degeneracy faces the domain-wall problem. GTE resolves it dynamically, using its own canonical coupling rather than by imposing an extra assumption. The resolution is testable: it predicts zero surviving defect network and a specific gravitational-wave ceiling, both falsifiable by current-generation experiments.

7.1 The domain-wall problem as a consistency test

The domain-wall problem is not a peripheral worry. For any theory that:

1. Has multiple degenerate discrete ground states, *and*
2. Claims to describe the early universe’s thermal history, *and*
3. Has no built-in mechanism to remove the walls,

a cosmological domain-wall catastrophe is *guaranteed*. The problem was first clearly articulated by Zel’dovich, Kobzarev, and Okun in 1974 (Zel’dovich, Kobzarev, Okun; see Further Reading). Any new physics beyond the Standard Model that introduces discrete symmetry breaking faces the same hurdle.

7.2 Three approaches to resolution

There are three broad strategies for solving the domain-wall problem:

Strategy	How it works	Cost
Inflation	Walls form before inflation, which dilutes them exponentially.	Requires a separate inflationary mechanism (absent in GTE, which predicts $r = 0$).
Explicit breaking	Add a term to the potential that explicitly breaks the discrete symmetry from the start.	Requires a new free parameter or a new theoretical input; not self-contained.
Bias annihilation	A small energy bias between vacua drives walls to collapse dynamically.	None, if the bias is already present in the canonical Lagrangian. <i>This is the GTE mechanism.</i>

GTE uses the third strategy, but without introducing any new input: the coupling $V_{\text{coupling}} = \varepsilon \varphi^2 (D_\mu \chi)^2$ is already present in the canonical Lagrangian, with $\varepsilon = 7/9$ fixed by MDL minimality. The resolution is not engineered; it is discovered.

7.3 The falsifiable nature of the prediction

Most cosmological defect scenarios are theoretically well-motivated but difficult to make precise: the size of the bias, the moment of annihilation, and the relic density depend on model parameters that can be tuned. In GTE, there are no tunable parameters. The wall tension is fixed by the BPS kink integral; the bias is fixed by the coupling coefficient $\varepsilon = 7/9$ and the color sector running; the annihilation temperature is $T_{\text{ann}} \approx T_G \approx 0.70$ GeV. The GW ceiling $\Omega_{\text{GW}} h^2 \lesssim 3 \times 10^{-47}$ follows.

This sharpness is what distinguishes a *prediction* from a *postdiction*. If LISA were to detect a stochastic gravitational-wave background at 0.02 Hz above the 3×10^{-47} ceiling, that specific number would be falsified — not merely challenged, but contradicted by a first-principles derivation.

7.4 Connection to the broader GTE picture

The domain-wall resolution is part of a cluster of zero-parameter cosmological predictions:

Observable	GTE prediction	Agreement
CMB spectral tilt n_s	$1 - \ln 2 / (2\pi^2) = 0.96488$	-0.005σ vs. Planck 2018
Baryon asymmetry η_B	6.109×10^{-10}	$+0.15\sigma$ vs. Planck 2018
$\mathbb{Z}_7$ defect relic	Zero surviving network	CMB+BBN No domain-wall relic observed (consistent)
GW ceiling (walls)	$\lesssim 3 \times 10^{-47}$ at 0.02 Hz	Not yet tested (far below LISA)
Dark energy eq. of state	$w_0 = -1$ exactly, $w_a = 0$	DESI Year-1 at 1.37σ (consistent)
Tensor-to-scalar ratio r	0 (no primordial GWs)	Current upper limit $r < 0.036$ (consistent)

All predictions carry zero free parameters. The domain-wall prediction fits naturally within a framework that derives — rather than postulates — the thermal history of the universe.

One-paragraph summary of the $\mathbb{Z}_7$ defect-cosmology result

The Φ_{MDL} field has seven equally-spaced degenerate ground states ($\mathbb{Z}_7$ vacua). When the universe cooled through the ordering crossover $T_G \approx 0.70$ GeV, independent causal regions chose different vacua (the Kibble mechanism), forming a domain-wall foam. Without a bias, this foam would have dominated the universe’s energy at recombination — catastrophic. The GTE Lagrangian’s canonical coupling term $\varepsilon\varphi^2(D_\mu\chi)^2$ ($\varepsilon = 7/9$, derived) provides exactly such a bias: the vacuum-label splitting $|\Delta V| \approx 3.8 \times 10^{-2}$ GeV⁴ drives the entire foam to annihilate within $\sim 10^{-23}$ s, 700 $\times$ before BBN, selecting the unique vacuum $k = 0$. This leaves zero surviving defect network — a parameter-free falsifiable prediction with a specific gravitational-wave ceiling $\Omega_{\text{GW}}h^2 \lesssim 3 \times 10^{-47}$ at $f \approx 0.02$ Hz.

A Quick Reference: Timeline and Scales

Epoch	Temperature	What happens
Reheating (post-bounce)	6.49×10^8 GeV	Universe filled with Φ_{MDL} excitations in thermal equilibrium; $\mathbb{Z}_7$ order is disordered
$\mathbb{Z}_7$ ordering crossover	$T_G \approx 0.70$ GeV	Kibble foam forms; all 7 vacua populated at $\approx 1/7$ each
Foam annihilation	$T_{\text{ann}} \approx T_G$	Coupling bias collapses foam within $\sim 5 \times 10^{-24}$ s
QCD crossover	~ 0.15 GeV	Quarks confine into hadrons
Big-Bang nucleosynthesis	~ 1 MeV	H, He, Li nuclei form; walls were gone $\sim 700\times$ earlier
Recombination	~ 0.25 eV	CMB photons decouple; domain-wall catastrophe would have occurred here without the bias
Today	2.7×10^{-4} eV	Zero surviving defect network

B Additional References

- **Full derivation of GTE defect cosmology:** P47 [1] — “GTE Cosmological Predictions” (Nova Spivack, 2026). doi.org/10.5281/zenodo.20465803
- **The Φ_{MDL} field and its Lagrangian:** P42 — “The Φ_{MDL} Field.” <https://doi.org/10.5281/zenodo.20417576>
- **The complete GTE monograph:** P48 [2] — “GTE: The Complete Theory” (Nova Spivack, 2026). doi.org/10.5281/zenodo.20560550
- **GTE polynomial tutorial:** “The GTE Polynomial: A Step-by-Step Cheat Sheet” (companion tutorial in this series).
- **Original domain-wall problem:** Ya.B. Zel’dovich, I.Yu. Kobzarev, L.B. Okun, Zh. Eksp. Teor. Fiz. **67**, 3 (1974).
- **Kibble mechanism:** T.W.B. Kibble, J. Phys. A **9**, 1387 (1976).

- **LISA gravitational-wave observatory:** P. Amaro-Seoane et al., arXiv:1702.00786 (2017).

Further Reading

Primary Sources

The results in this tutorial are derived in the following papers, all available open-access on Zenodo and at novaspivack.com/research:

- **P47 — GTE Cosmological Predictions** (defect cosmology, relic constraints):
doi.org/10.5281/zenodo.20465803
- **P42 — The Φ_{MDL} Field** (kink/domain-wall spectrum):
doi.org/10.5281/zenodo.20417576
- **P48 — The Complete GTE Framework** (main synthesis monograph):
doi.org/10.5281/zenodo.20560550

Lean 4 machine proofs: github.com/novaspivack/ugp-lean

References

- [1] Nova Spivack. Cosmological predictions of the GTE/ Φ_{MDL} framework. Preprint (Zenodo), 2026. URL <https://doi.org/10.5281/zenodo.20465803>. <https://doi.org/10.5281/zenodo.20465803>. UGP Physics series, P47.
- [2] Nova Spivack. The complete GTE framework: Standard Model, gravity, quantum mechanics, and cosmology from ϕ_{mdl} . Zenodo, 2026. URL <https://doi.org/10.5281/zenodo.20560550>. <https://doi.org/10.5281/zenodo.20560550>. Preprint.